

PROJECT ABSTRACT
PROJECTNEST – REAL-TIME CODE
COLLABORATION SYSTEM

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1. Overview

This project introduces a Real-Time Code Collaboration System designed for programmers who work together on shared coding projects across different systems. The platform allows users to create both public and private projects, manage team members, and collaboratively edit files with instant synchronization. Built using Python, Django, and Django Channels, it provides a modern coding environment where any update made by one user is immediately visible to others without reloading the page. The system aims to offer a simple and user-friendly alternative to complex version-control tools, making real-time collaboration accessible to students, beginners, and small teams.

2. Objectives and Scope

The objective of this project is to build a secure and interactive platform that supports real-time collaborative coding, allowing users to create projects, manage members, work on files through an in-browser editor, and experience instant synchronization of changes. The scope includes essential functionalities such as user authentication, project creation, file management, activity tracking, real-time updates, and access to public projects. While not intended to replace advanced systems like GitHub, the platform fully supports the needs of small development groups and academic projects.

3. Modules

3.1 Admin Module

This module uses Django's built-in admin panel to manage the platform at a system-wide level. Admins can view, add, edit, or delete users, projects, files, and roles. They oversee user registrations, monitor projects for misuse, manage public content, and maintain security. By leveraging Django Admin, the module provides full administrative control without extra coding, ensuring smooth operation and platform integrity.

3.2 Project Owner Module

The project owner is responsible for creating and leading projects. They can set project visibility as public or private, invite contributors, approve or reject join requests, configure member permissions, and monitor project activities. This module ensures that every project has structured leadership and controlled access to team members.

3.3 Contributor Module

Contributors actively participate in projects by editing, saving, and collaborating on files in real time. They do not have ownership privileges but play a key role in building and updating the project. Through this module, contributors can work together efficiently while staying informed via activity logs and notifications.

3.4 Viewer Module

Viewers have access to projects for observation purposes only. They can browse project files, review existing code, and follow activity logs but cannot make edits or delete files. This module allows users to learn from projects and monitor progress without interfering with ongoing work.

3.5 Public User Module

Public users explore the platform without being part of any private projects. They can browse publicly shared projects, view details, and download files if permitted. While they cannot contribute directly, they can request access to private projects or follow public projects for learning. This module promotes community engagement and knowledge sharing.

4. Conclusion

The Real-Time Code Collaboration System successfully demonstrates the practical use of modern web technologies to support seamless multi-user coding environments. Through its structured modules, the platform provides an efficient, secure, and user-friendly solution for collaborative programming. It highlights core concepts of Django, Django Channels, and real-time communication, making it a valuable tool for students, beginners, and small development teams working on shared coding projects.